

Advanced JAVA

Unit-1

Lab Programs

1. Write a Java program to demonstrate the concept of classes and objects. Create a class named Student with data members such as student name and student age. Create an object of the class and display the student details using a method.

```
1 // Program to demonstrate Classes and Objects
2
3 class Student {
4     // Data members
5     String studentName;
6     int studentAge;
7
8     // Method to display student details
9     void displayDetails() {
10         System.out.println("Student Name: " + studentName);
11         System.out.println("Student Age: " + studentAge);
12     }
13 }
14
15 public class Main {
16     public static void main(String[] args) {
17         // Creating an object of Student class
18         Student s1 = new Student();
19
20         // Assigning values to data members
21         s1.studentName = "Sushaina";
22         s1.studentAge = 20;
23
24         // Calling the method to display details
25         s1.displayDetails();
26     }
```

Output

```
Student Name: Sushaina
Student Age: 20
```

```
=== Code Execution Successful ===
```

2. Write a Java program to demonstrate the concepts of inheritance and polymorphism. Create a parent class named **Animal** and a child class named **Dog**. Use method overriding to achieve runtime polymorphism and display the behavior of both classes.

Main.java



Share

Run

```
1 // Program to demonstrate Inheritance and Runtime Polymorphism
2
3 // Parent class
4 class Animal {
5     void sound() {
6         System.out.println("Animal makes a sound.");
7     }
8 }
9
10 // Child class
11 class Dog extends Animal {
12     @Override
13     void sound() {
14         System.out.println("Dog barks.");
15     }
16 }
17
18 // Main class
19 public class Main {
20     public static void main(String[] args) {
21
22         // Parent class object
23         Animal a1 = new Animal();
24         a1.sound();
25
26         // Runtime Polymorphism
27
28         Animal a2 = new Dog();
29         a2.sound();
30     }
```

Output

Animal makes a sound.
Dog barks.

=== Code Execution Successful ===

